FEBAUTO Racing

Release 2: starting fresh

For	anyone who has not written driver code yet, with or without the old devkit
Time	about 20 minutes, most of it a download
Result	the simulator driving, a ROS 2 terminal, rviz in a browser tab

Formula Electric at Berkeley
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https://github.com/FEBAutonomous/feb-racing
https://febautonomous.github.io/feb-racing/

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1 Before you start

Install these three, once:

1. **Docker.** Mac: Docker Desktop or OrbStack. Windows: Docker Desktop, and in its settings turn on *WSL integration* for your Ubuntu. Linux: `sudo apt install docker.io` and add yourself to the `docker` group, then log out and in.
2. **VS Code** with the *Dev Containers* extension. Cursor works the same way.
3. **Foxglove** from `https://foxglove.dev/download`. Optional, the browser desktop covers most of it.

Windows: everything below happens in a **WSL 2 Ubuntu terminal**, not PowerShell. If you do not have one: `wsl -install -d Ubuntu` in PowerShell, reboot, open "Ubuntu" from the Start menu.

2 Had the old devkit? Delete it

Release 2 rewrote the repository, so the old clone cannot be updated. It holds nothing you need if you never wrote code in its `stack/` folder. Remove it:

```
rm -rf ~/feb-racing
```

If it lives somewhere else, delete it from there. Skip this section if you never had it.

3 Setup

```
mkdir -p ~/FEB && cd ~/FEB
git clone git@github.com:FEBAutonomous/feb-racing.git
cd feb-racing
./feb setup --name "Your Name"
```

What happens: the launcher pulls the devkit image (a few GB, once), downloads the simulator app for your OS, and creates *your box*, a container with ROS 2 Humble, the house nodes and a desktop, with your whole `~/FEB` folder mounted inside it. The box is persistent: it survives reboots, keeps what you `pip install`, and builds incrementally.

Your name is how your results appear on the scoreboard. Ask Pranav to add it to `submissions.yaml` with your GitHub login.

No `git@github.com` access yet? Use `https://github.com/FEBAutonomous/feb-racing.git` in the clone line; everything else is the same.

4 First drive

```
./feb up
```

The simulator window opens. Arrow keys drive. Close the window, or Ctrl-C in the terminal, to stop. Mac, first time only: if macOS refuses to open the simulator, right-click the app in `~/Library/Application Support/feb-sim/app` and choose Open.

To watch the starter driver race instead of driving yourself:

```
./feb up --driver feb_driver
```

5 Look inside: a terminal, rviz in the browser, Foxglove

Command	Gives you
`./feb shell`	a terminal inside the box; `ros2 topic list`, `ros2 topic echo /scan`
`./feb shell --gui`	the same, plus the box's desktop in a browser tab at http://localhost:6080
`./feb code`	VS Code opened inside the box, with the ROS extensions

rviz in the browser. With `./feb up` running in one terminal, open a second one:

```
./feb shell --gui
rviz2 &
```

A browser tab opens with a grey desktop; rviz appears on it. In rviz: *Fixed Frame* to `odom`, then *Add*, *By topic*, `/scan` (LaserScan) and `/odom` (Odometry). Set the LaserScan style to Points, size 4. If a topic shows nothing, set its *Reliability Policy* to Best Effort. `rqt` and `turtlesim` land on the same desktop.

Foxglove app. Open connection, Foxglove WebSocket, `ws://localhost:8765`. Add a 3D panel, turn on `/scan`; the bridge is already running whenever `feb up` is.

6 Day to day

Command	Does
`./feb update`	pulls the repo, a newer image if there is one, a newer simulator. Run it when told there is a release.
`./feb stop`	stops a run that is stuck
`./feb reset`	throws the box away and makes a new one; your files are untouched
`./feb practice --driver my_driver`	scores a driver on the qualifying tracks
`./feb submit`	posts clean runs to the scoreboard and your code to the team repo

You never re-clone again. `./feb update` is the whole upgrade path from here on.

Writing a driver, scoring, the dead-man and the car are in `member_guide.pdf`, next to this file.

7 When something is wrong

It says	Do
`Cannot connect to the Docker daemon`	Docker Desktop is not running; open it and wait for the whale icon to settle
`this clone sits directly in /home/...`	the clone must be inside a folder of its own: `mkdir ~/FEB && mv ~/feb-racing ~/FEB/`
`unauthorized` while pulling the image	you are logged in to `ghcr.io` with a stale token: `docker logout ghcr.io`, run setup again
the simulator window never appears	`./feb logs`; on Windows make sure `~/FEB` is inside WSL, not on `/mnt/c`
the browser desktop will not connect	reload the tab; still nothing, `./feb reset`